The Direct and Indirect Effects of Genetics and Education

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Abstract

Social science genetics has shown genes matter for education and labour market outcomes, but has little credible evidence on the economic mechanisms beneath these findings. I use plausibly random deviation from parents as a natural experiment for the Education PolyGenic Index (Ed PGI) to estimate the causal effect of genes associated with years of education, with data from the UK Biobank. I then decompose the total genetic effects into a direct genetic effect and an indirect education pathway using a causal mediation framework that allows for selection into education. A credible mediation analysis requires causal estimates of the returns to education, which I import from a meta-analysis of over one hundred estimates from the economics literature. The Ed PGI increases labour market earnings primarily through higher education attendance, with at least 60% of the total effect operating through education. The remainder is consistent with small direct genetic effects; education-linked genes are more about returns to higher education, and less about direct genetic effects. These findings reframe the usual focus in social science genetics by shifting from asking if genes matter to how they matter — not as deterministic forces, but as factors that influence human capital decisions within institutional constraints. By using causal methods that account for individual choice in education, rather than assuming genetic determinism, this research provides a richer understanding of how biological inheritance, human capital decisions, and institutional structures interact to shape labour market inequality.

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JEL Codes: D31, D91, I24, J24, Z00.

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> The difference of natural talents in different men, is, in reality, much less than we are aware of [...]. The difference between the most dissimilar characters, between a philosopher and a common street porter, seems to arise not so much from nature, as from habit, custom, and education.
>
> — *Smith (1776), The Wealth of Nations, Volume 1 p. 16.*

0.1 Writing Plan

The introduction formula https://blogs.ubc.ca/khead/research/research-advice/formula**.**

Note: a major concern for the Ed PGI previous papers is that they have little economic insight. This paper is framed as changing that, investigating the economic mechanisms underlying the Ed PGI causal effects, giving economic credence where before there was merely concern.

Papageorge & Thom (2020) document how the Ed PGI predicts earnings, even when controlling for education, and posit that the index contains genes that affect skills related to financial literacy.

Quote for the Mendelian independent first-stage, "Cannot choose who your parents are, and you also cannot choose how much you inherit from them."

1 Introduction

This paper proceeds as follows. Section 2 summarises the UKB data, and explains how the Ed PGI is constructed. Section 3 shows a causal research design, based on Mendelian independent inheritance, for estimating the causal effects of the Ed PGI; Section 4 shows estimates of the causal effects of the Ed PGI, on education and labour market outcomes. Section 5 gives a conceptual framework for estimating the direct and indirect effects of genetics and education, in this observational setting; Section 6 gives the empirical results. Section 8 concludes.

2 Data and Genetic Context

This paper draws on data from the UK Biobank, a large-scale cohort study of the British population with both rich socioeconomic measures and detailed genetic data. I construct a

Education PolyGenic Index (Ed PGI) summarising each participant's genetic predisposition toward educational attainment, and impute parental PGI values using genetic data from siblings. Together, these data enable a within-family research design that separates the role of inherited genetic factors from the shared family environment in shaping socioeconomic outcomes.

2.1 UK Biobank (UKB)

The UK Biobank (UKB) is a large-scale prospective cohort study comprising approximately half a million members of the British public, recruited between 2006 and 2010. Letters were sent to a random sample of addresses in the National Health Service registry, and the final UKB data represent voluntary respondents.[1] Participants, aged 40-69 years at recruitment, underwent comprehensive baseline assessments, were asked many questions for socioeconomic information either by interview, automated survey answers, or follow-up sessions. Key socioeconomic variables include educational attainment (measured in education qualifications), household income brackets, employment status, and industry of employment. See Sudlow et al. (2015) for a full description of the UKB organisation and data.

The UKB collected detailed information on educational qualifications, including completion of primary school, GCSEs (typically completed at age 16), A-Levels (completed at age 18), and completion of an higher education degree. I convert these qualifications into education years based on each participant's highest reported qualification, following the International Standard Classification of Education.[2] While the UKB collected household income data, this information was recorded only in categorical bins, limiting its utility for continuous analysis. Following Kweon et al. (2025), I construct an individual-level income measure using participants' 4-digit Standard Occupational Classification (SOC) codes. Specifically, I impute occupation-based wages by estimating hourly earnings for each SOC category using data from the British Annual Survey of Hours and Earnings, conditional on age, gender, and year. This approach provides a more granular and continuous measure of economic status than the categorical household income variable (see Appendix A.1 for details on the imputation procedure). I create the annual occupation income by multiplying this by the number of

[1] The UKB may not be representative of the entire UK population, as it only represents willing respondents. As such, the findings of this paper should only be taken as internally valid among the UKB respondents, and the sibling analysis sample described.

[2] As noted by Okbay et al. (2022), British vocational degrees are inconsistently coded by the International Standard Classification of Education. I follow the literature by assigning education years as the age they reported leaving full-time education minus five, among those with a vocational degree as their highest qualification.

Table 1: Descriptive Statistics, UKB.

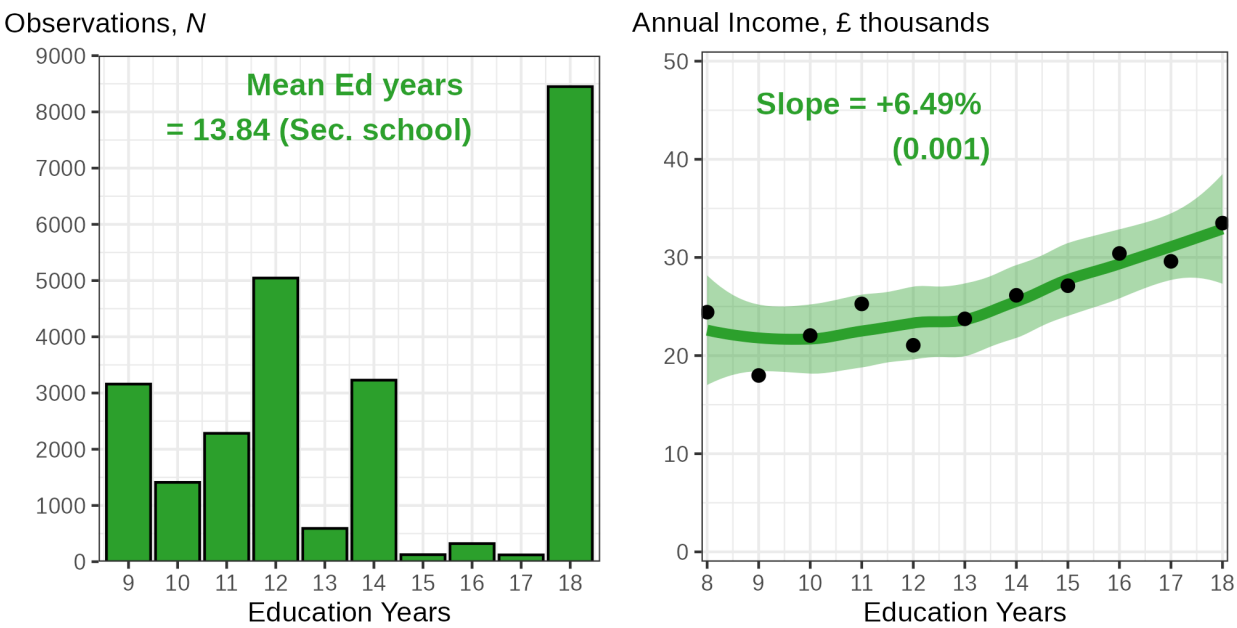
	Analysis sample		Entire sample	
	Mean	SD	Mean	SD
Demographics:				
Male	0.44	0.50	0.46	0.50
Age	55.04	7.05	56.55	8.09
Lives in city	0.86	0.35	0.85	0.35
Ethnicity = European	1.00	0.00	0.82	0.38
Any siblings also in UKB?	1.00	0.00	0.08	0.28
Count of siblings in UKB	1.09	0.32	0.09	0.32
Genetic Measures:				
Ed PGI	0.04	1.00	0.00	1.00
Ed PGI, imputed parental mean	-0.02	0.75		
Other PGI: ADHD	-0.02	1.00	-0.00	1.00
Other PGI: Asthma	-0.01	1.00	0.00	1.00
Other PGI: Bipolar	-0.01	1.00	-0.00	1.00
Other PGI: BMI	-0.02	0.99	0.00	1.00
Other PGI: Diabetes (type 2)	-0.02	1.00	0.00	1.00
Other PGI: Height	0.00	1.01	-0.00	1.00
Other PGI: Schizophrenia	-0.04	1.00	0.00	1.00
Education:				
Education years	13.84	3.36	13.72	3.48
Age left education	16.67	2.47	16.51	3.04
Qualification, University degree	0.33	0.47	0.33	0.47
Qualification, A-Levels	0.46	0.50	0.45	0.50
Qualification, GCSEs	0.76	0.43	0.71	0.45
Qualification, Professional degree	0.32	0.47	0.29	0.45
Qualification, Vocational degree	0.23	0.42	0.19	0.39
Qualification, No official qualifications	0.13	0.34	0.17	0.38
Labour Market Outcomes:				
Occupation hourly wage, £	18.63	8.00	19.23	8.31
Occupation annual income, thousands £	26.33	16.57	27.83	17.48
Average hours worked, per week	34.73	12.58	35.31	12.70
Household income, $< £18k$	0.13	0.33	0.20	0.40
Household income, $£18 - 31k$	0.23	0.42	0.22	0.41
Household income, $£31 - 52k$	0.28	0.45	0.22	0.42
Household income, $£52 - 100k$	0.22	0.41	0.17	0.38
Household income, $£100k <$	0.05	0.22	0.05	0.21
Household income, thousands £ (midpoint imputed)	44.00	26.23	39.97	27.18
Observations	24,743		496,004	

Note: This table shows the Mean and Standard Deviation (SD) of UKB variables, comparing the analysis subsample and the entire UKB sample. The analysis sample refers to individuals with at least one sibling in the UKB data, of European ancestry. All genetic variables are only non-missing among those who report European ancestry.

hours they reported working, times 52 weeks in the year. For example, this procedure would give a female participant, age 58 in 2010 with SOC 2314 (i.e., a secondary education teacher by profession), an occupation hourly wage of £29.20 and an annual income of £68,476 for her 45 hours worked each week (adjusted to 2024 £GBP by CPI-H).

Figure 1: Distribution of UKB Education Measures.

(a) Education Years. **(b)** Education Years + Income.



Note: This figure shows the marginal distribution of education years in the UKB analysis sample, and the correlation between years of education and occupation coded income.

For the purposes of this study, the primary analysis sample is restricted to those with at least one family member, sibling or parent, also enrolled in the UKB, and non-missing data in occupation code plus education qualifications. The UKB does not personally identify participants, nor their family members, so I follow the genetic literature by identifying family clusters by measures of genetic similarity.[3] The presence of multiple family members within the dataset provides a unique opportunity to control for shared genetic factors that might otherwise bias estimates in standard cross-sectional analyses.

The analysis sample is also restricted to individuals who report to have primarily European

[3]The design of the UKB allows for a larger number of family clusters than would be expected from a random sample of the British population. This is because study enrolment letters were sent to households, and the household head (or to whomever the letter was addressed) could optionally enrol their family members in addition to themselves. Appendix A.2 describes the family cluster identification procedure in detail, which has been used in previous social science genetic studies (see e.g., Muslimova et al. 2025).

ancestry. The genetic measures used in this paper have generally been constructing from primarily white, European descended populations, and have no predictive power out of sample, so are only relevant in this subsample.[4] The only notable differences between the analysis subsample and the entire UKB participant sample is that the analysis sample has more siblings, and is entirely European descended; no other variables are significantly different for the analysis subsample. Table 1 summarises the UKB data, for the analysis sample and among the entire UKB sample. Figure 1 summarises the correlation between education years and occupation coded income.

2.2 Education PolyGenic Index (Ed PGI)

UKB participants also provided saliva samples that were subsequently genotyped using high-density arrays containing over 800,000 genetic markers, producing standardised genetic data files for statistical analysis of the relationship between genetics and social outcomes.

While genetic data provide rich information about individual variation, the relationship between specific genes and social outcomes like education is extraordinarily complex. The biological pathways from DNA to observable traits remain largely opaque — the exact relationship between a single gene, its resulting protein (or its moderating role), and the effects of those proteins are only known for a small number of genes in the human genome. For example, a single gene (and its recessive variant) is known to lead to a protein which does not break down the by-products of alcohol digestion; individuals with this gene generally do not drink alcohol at all because of the simple limit on digestion (Millwood et al. 2023). Complex social behaviours, on the other hand, have no single genes with a known biological pathway, so cannot be summarised in a single gene, nor single biological pathway. Instead, complex social behaviours are a result on thousands of genes, and their interactions with the environment — they are polygenic.

Rather than attempting to trace all of these intricate biological mechanisms, researchers have developed PolyGenic Indices (PGIs) as dimensional reduction tools that summarise the statistical relationship between thousands of genetic variants and outcomes of interest. This aggregation approach is particularly valuable because individual genetic variants typically have tiny effects on complex traits like educational attainment — often explaining less than 0.01% of the variation. However, by combining information across hundreds of thousands of

[4] It is an unfortunate fact that systematic genetic measures, such as those considered in this paper, have only been constructed from Biobank samples where the vast majority are white and/or European descended; see Martin et al. (2017) for a further discussion. Note that this implies this paper has nothing to say about race; see Subsection 2.4.

variants, each contributing a small effect, PGIs can capture meaningful portions of the genetic contribution to these outcomes. These indices are fundamentally correlational measures: they capture statistical associations between genetic variants and outcomes observed in large samples, not well understood biological pathways, and without a clear case for causal relationships. The weights used to construct PGIs reflect population-level correlations that may arise through various channels, including biological effects, gene-environment correlations, and population stratification.

The Education PGI (Ed PGI) is a summary measure that aggregates genetic information across thousands of genetic variants into a single index predictive of educational attainment. The construction of the Ed PGI relies on weights derived from a Genome-Wide Association Study (GWAS), which analyse the correlation between genetic variants and education years. In a GWAS, researchers test the correlation between each Single Nucleotide Polymorphism (SNP) — a one-letter change in the DNA sequence at a specific location — and the outcome of interest,[5] in this case education years. I use the weights calculated in the GWAS conducted by Okbay et al. (2022), which used a sample of over 3 million individuals of European ancestry, primarily from 23&Me data.

The Ed PGI is defined as the weighted sum of whether individual i has the relevant SNP at each DNA location j,

$$Z_i = \sum_{j=1}^{J} w_j x_{i,j}$$

where w_j are the Ed PGI weights (provided in the supplementary data to Okbay et al. 2022), and $x_{i,j}$ is an indicator of whether individual i has the relevant SNP at DNA location j.[6] The Okbay et al. (2022) GWAS found that over 8,000 individual SNPs were significant in predicting education years, after passing a battery of biostatistical tests and finding no evidence for significant interactions. The resulting Ed PGI is standardised to have mean zero and standard deviation one within the entire sample, making it interpretable as the number of standard deviations from the population mean. The Ed PGI captures approximately 12–15% of the variation in educational attainment, making it one of the most predictive PGIs for any complex behavioural trait to date.

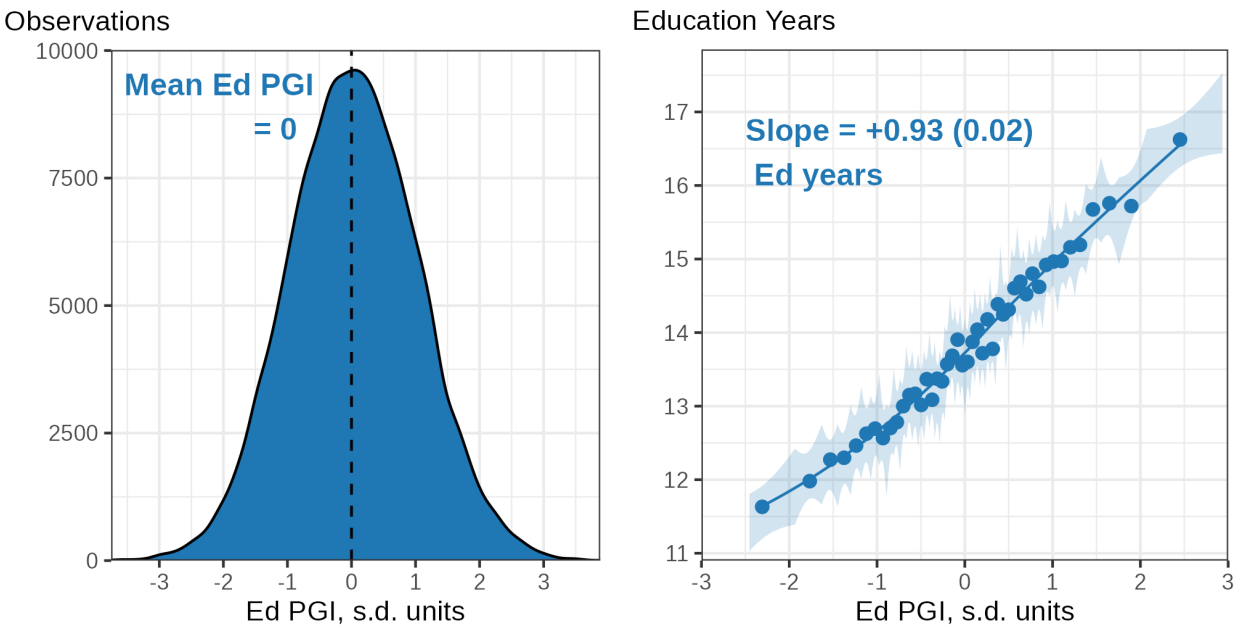
If education had no relationship with SNP variants, then we would expect the correlation

[5] See Uffelmann et al. (2021) for a full explanation of how a GWAS works, and the approach to calculating PGIs.

[6] Every person has two copies of each chromosome, so that $x_{i,j}$ takes values 0, 1, 2 corresponding to how many copies of the SNP individual i has inherited. The genetics literature has generally accepted that complex social behaviours are explained by additive effects (see Hill et al. 2008), and Okbay et al. (2022) found little evidence for dominance in the Ed PGI SNPs, so that the Ed PGI makes sense as a weighted sum.

Figure 2: Distribution of Education PolyGenic Index (Ed PGI).

(a) Ed PGI. **(b)** Ed PGI + Education Years, Binscatter.



Note: This figure shows the marginal distribution of Ed PGI and education years in the UKB analysis sample. Ed PGI is normalised to have mean 0 and standard deviation 1 among the entire UKB sample; it is roughly normally distributed regardless of this normalisation.

between the Ed PGI and education years was indistinguishable from zero; it is, however, highly correlated with education years. A one standard deviation increase in the Ed PGI is associated with 0.93 extra education years (standard error 0.02), with an R^2 value of 0.10 for the raw regression of education years on Ed PGI. It is roughly normally distributed, with standard deviation normalised to 1. This distribution is not surprising, as the Ed PGI is the weighted sum of a few thousand binary variables, so we should expect convergence towards a normal distribution among the population. The main take-away is that the correlation between a summary measure of genetic values is strong, and has meaningful predictive power for education years.

The Ed PGI is not the only PGI that can be constructed. Following the same methodology, I construct additional PGIs for six health-related traits that may confound the relationship between genetics and socioeconomic outcomes: asthma, bipolar disorder, Body Mass Index (BMI), height, schizophrenia, and type 2 diabetes. Each PGI is calculated using the same approach as the Ed PGI — taking weights from a published GWAS, conducted on independent populations, and computing weighted sums of the relevant genetic variants in the UKB data. These other PGIs serve as control variables in my analysis for two reasons. First, genetic

variants often exhibit pleiotropy, meaning a single variant can influence multiple traits simultaneously. For instance, genetic variants associated with education might also affect other factors (e.g., mental health conditions), which could independently influence labour market outcomes. Second, systematic differences in gene frequencies across ancestry groups can create spurious correlations between any PGI and outcomes if health conditions vary across these groups for unrelated reasons.

2.3 Imputing Parental PGIs from Siblings

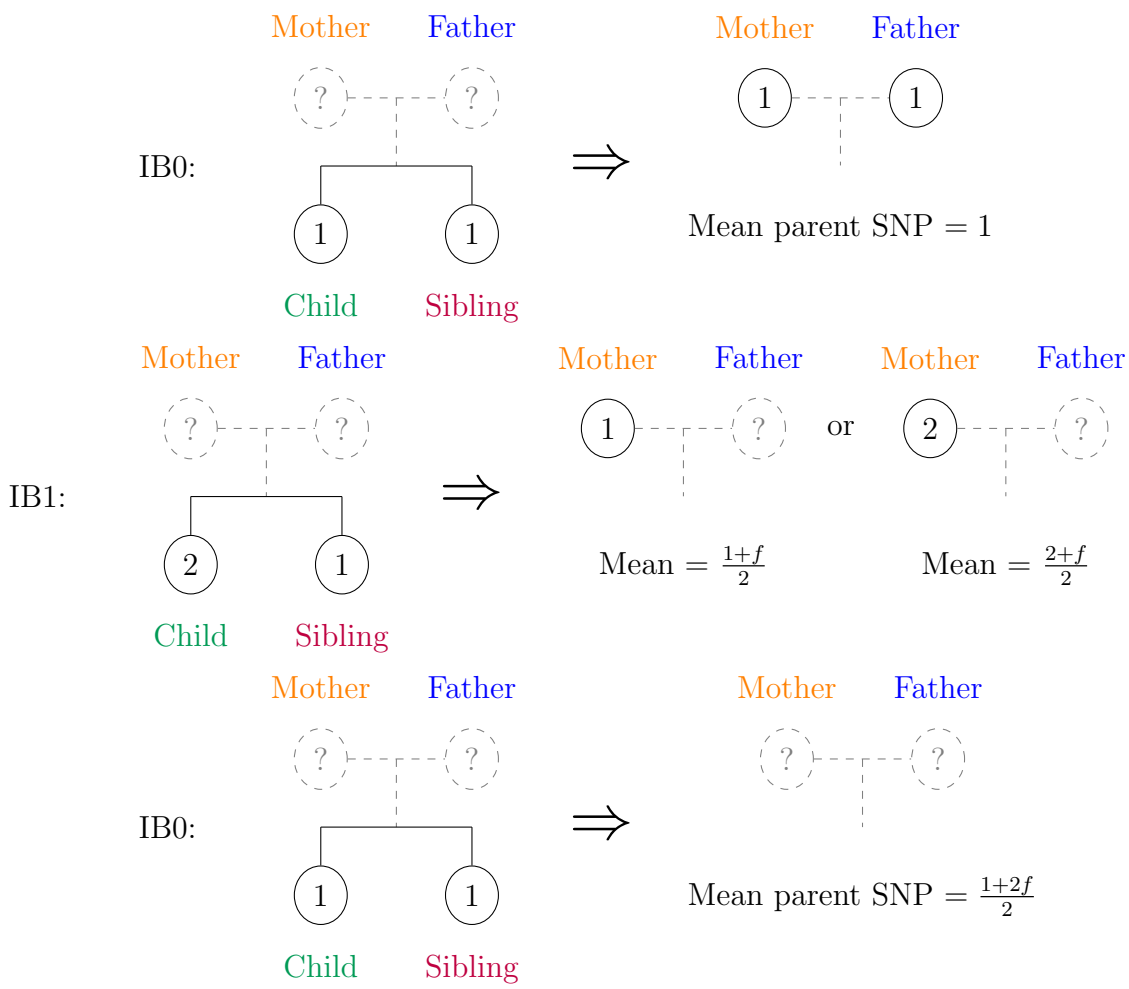
It would be ideal if the UKB contained genetic information for the parents of every UKB participant. While there are 6,239 participants with a parent also included in the UKB, there are only 72 with both parents. To overcome this missing data problem, I use siblings' genetic data to impute the mean parental PGIs from the sample of 24,743 participants with at least one sibling also in the UKB — the sibling analysis sample described above.

I use the Young et al. (2022) procedure to impute mean parental PGI values using genetic data on participants' siblings. The key insight is that since siblings both inherit their DNA from the same two parents, observing two siblings' genotypes allows us to partially reconstruct the parental genotypes they were drawn from. At each SNP, the imputation proceeds in two steps. First, the procedure classifies each sibling pair as being in one of three identity-by-descent (IBD) states using a hidden Markov model that exploits information across all SNPs: IBD0, in which the siblings share no alleles inherited from the same parent; IBD1, in which they share one such allele; and IBD2, in which they share both. This IBD classification has been shown to be correct 99.65% of the time in validation samples (Young et al. 2022). Second, given the inferred IBD state, the procedure imputes the sum of the two parental genotypes at that SNP — which is the quantity needed to construct a parental PGI — by summing the directly observed parental alleles and replacing any unobserved alleles with the population allele frequency $f \in [0, 1]$.

Figure 3 illustrates this logic for the case where both the child and their sibling have observed SNP value of 1 (one copy of the reference allele). In IBD0, the siblings share no alleles, so all four parental alleles are directly observed across the two children, and the parental sum is known exactly as 2. In IBD1, one allele is shared between the siblings, leaving one parental allele unobserved; depending on whether the shared allele is the reference allele or not, the imputed parental sum is either $1 + f$ or $2 + f$, depending on value of the inherited shared allele.[7] In IBD2, both siblings inherited the same two alleles, so only two of the

[7] Phased haplotype data from neighbouring SNPs used to resolve this ambiguity.

Figure 3: Imputing One SNP for Parents from Sibling Data.



Note: This figure illustrates the logic of imputing the sum of parental genotypes from sibling data at a single SNP, following Young et al. (2022). At each SNP, individuals have a genotype of 0, 1, or 2, representing the number of copies of a reference allele inherited from their parents; this figure shows the case where both the child and their sibling have genotype 1, across the three possible identity-by-descent (IBD) states. In IBD0, the siblings share no alleles by descent, so all four parental alleles are directly observed and the parental sum is known exactly as 2. In IBD1, the siblings share one allele by descent, leaving one parental allele unobserved; the imputed parental sum is $1 + f$ or $2 + f$, depending on whether the shared allele is the reference allele, where f denotes the population frequency of the reference allele. In IBD2, the siblings share both alleles by descent, so only two parental alleles are observed and the imputed parental sum is $1 + 2f$. In all cases, unobserved parental alleles are replaced by the population allele frequency f.

four parental alleles are observed (summing to 1), and the two unobserved alleles are each imputed as f, giving sum of parents $1+2f$. In all cases, the imputed value is the conditional expectation of the parental sum given the observed sibling genotypes and IBD state, where unobserved alleles are replaced by the population frequency. Averaging across the three IBD states — which occur with probabilities 1/4, 1/2, and 1/4 respectively — siblings jointly

reveal on average three of the four parental alleles, meaning approximately 75% of parental genetic information is directly observed rather than imputed.[8]

2.4 Ethical Considerations

The genetic measures in this study should not be misinterpreted as evidence of genetic determinism. Having a higher Ed PGI does not destine someone to higher education or income; it represents a statistical tendency, not fate. These tendencies are fundamentally shaped by social policies, institutional structures, and cultural contexts that vary across time and place. Just as eyesight correcting glasses can eliminate the practical consequences of genetic variation in eyesight, education and economic policies can modify or eliminate the consequences of genetic variation in traits relevant to socioeconomic outcomes.

Furthermore, this research provides no support for ideas about "natural" social hierarchies or genetic justifications for inequality. In addition, genetic associations identified within one ancestry group cannot be meaningfully compared across groups due to different genetic architectures, environmental contexts, and historical experiences. Any claim using the measures considered in this paper for inference across racial groups, or different racial groups, would be both invalid and ethically deficient. The appropriate interpretation of these results is that they highlight how both social and genetic factors influence socioeconomic outcomes, which should inspire greater solidarity and support for policies that promote equality of opportunity rather than acceptance of inequality as "natural" or inevitable.

3 Random Variation in Genetic Inheritance

Genetic variants are not randomly assigned, which means that simple correlations between the Ed PGI and observed outcomes cannot straightforwardly be interpreted as causal. Genetic variation arises from inheritance: at conception, individuals inherit 23 chromosomes from each parent, obtaining half of their genetic variants from their mother and half from their father. Because parents transmit genetic material to their children, those who inherit a higher Ed PGI typically also inherit familial environments shaped by parents who themselves possess a higher Ed PGI. Consequently, correlations between an individual's Ed PGI and outcomes may reflect not only direct genetic effects but also indirect effects mediated through

[8] See Young et al. (2022) for further reasoning on the imputation performance relative to observed parents in the UKB, and Section 7 for a discussion on validating this design in an alternative data source which observes both siblings and parents.

family environment, making these correlations confounded.

However, while genetic inheritance generates systematic associations between genes and environments, it also includes a random component. Specifically, Mendelian inheritance ensures that there is random variation in the genetic variants a child inherits — conditional on parental genetics. I use this random variation to estimate causal effects of the Ed PGI.

3.1 Mendelian Independent Inheritance

I use variation away from parental values to estimate causal effects of having a higher versus lower Ed PGI. This approach leverages the fundamental randomness in genetic inheritance, first described by Mendel's laws (Mendel 1865).

Each parent has 0, 1, or 2 copies of each genetic variant — corresponding to presence on each of their chromosomes. When a father and mother's sex cells are formed, exactly one of the two copies is sorted into each reproductive cell, and is then inherited by their resulting child. A parent with one copy therefore passes it on half the time, whereas a parent with 0 or 2 copies transmits their genetic variant with certainty. At conception, the child receives one draw from the mother and one from the father, so the child's copy-count is simply the sum of two independent draws.

Write $x_{i,j}^{\text{Child}}$ for the number of copies of variant j that individual i has (0, 1, or 2), and $x_{m(i),j}^{\text{Mother}}, x_{f(i),j}^{\text{Father}}$ for their respective parents $m(i), f(i)$. If the mother has two copies of variant j then they bequeath one copy to their child with certainty; if the father has one, then they bequeath one copy with probability $\frac{1}{2}$; if no copies, then they bequeath none with certainty. The child then receives their other copy from their father, by the same process, giving the following representation in expectation,

$$\mathbb{E}\left[x_{i,j}^{\text{Child}} \,\middle|\, x_{m(i),j}^{\text{Mother}}, x_{f(i),j}^{\text{Father}}\right] = \frac{1}{2}\left(x_{m(i),j}^{\text{Mother}} + x_{f(i),j}^{\text{Father}}\right).$$

After conception, the child has inherited genetic variants from both parents — one of the 4 possible discrete realisations, not the expected value. This result came about purely from random chance, conditional on at least one of parent having exactly one copy of the SNP, $x_{m(i),j}^{\text{Mother}} = 1$ and/or $x_{f(i),j}^{\text{Father}} = 1$.[9]

[9]Note both parents having 0 or 2 copies leaves $x_{i',j'}^{\text{Random}} = 0$, so that individual i' inherited no random variation in variant j'.

3.2 Mendelian Independent Inheritance in the Ed PGI

Moving from a single variant to the PGI simply means summing across the same random-segregation for all the variants. The Ed PGI is constructed by summing the 0, 1, or 2 copy indicators across the $J \approx 8,000$ different SNPs, each multiplied by the externally estimated weights $w_{j'}$ that capture how strongly each SNP predicts education years.

$$Z_i^{\text{Child}} = \sum_{j=1}^{J} w_j x_{i,j}^{\text{Child}},$$

and similarly for child i's parents, to give $Z_{m(i)}^{\text{Mother}}$ and $Z_{f(i)}^{\text{Father}}$ from summing across binary values $x_{m(i),j}^{\text{Mother}}$ and $x_{f(i),j}^{\text{Father}}$, respectively.

Because every variant copy enters the child's genome through an independent random draw (conditional on parental values), the child's Ed PGI is the weighted sum of thousands of such draws. This means the Ed PGI can be decomposed into the component that was inherited as a result of a random draws from parent(s) who had 1 copy of the variant, and the certainty component from draws where parent(s) had 0 or 2 copies.

$$Z_i^{\text{Child}} = \mathbb{E}\left[Z_i^{\text{Child}} \,\middle|\, Z_{m(i)}^{\text{Mother}}, Z_{f(i)}^{\text{Father}}\right] + u_i, \tag{1}$$

where u_i is a mean zero random error term arising from Mendelian independent assortment. The only determining factor for u_i is the randomness of inheriting exact counts of SNPs $x_{i,j}^{\text{Child}}$ from their parents, relative to what they were expected to inherit, $\frac{1}{2}\left(x_{m(i),j}^{\text{Mother}} + x_{f(i),j}^{\text{Father}}\right)$.

Ideally, the expected inheritance is constructed from observed parental Ed PGI values, $Z_{m(i)}^{\text{Mother}}, Z_{f(i)}^{\text{Father}}$. However, parental PGI values are not observed for a large enough sample in the UKB for statistical inference. Thus, I follow Young et al. (2022) and leverage siblings' genetic data to impute parental PGI values. The expected Ed PGI inheritance, therefore, is calculated using the same within-family variation that early genetic studies did by examining within family differences. An individual's Ed PGI is drawn from the same distribution as their sibling, because they had the same parents; in absence of observing both parents directly, this approach uses the sibling to infer this distribution.

This approach is equivalent to using observed parents because of the simple fact that children and parents are two independent draws from the parental genetic distribution. Write $Z_{s(i)}^{\text{Sibling}}$ for child i's observed sibling Ed PGI value, and $Z_{i,s(i)}^{\text{Parental}} = \frac{1}{2}\left(Z_i^{\text{Child}} + Z_{s(i)}^{\text{Sibling}}\right)$ the imputed parental value. In expectation, a child and their observed sibling's Ed PGI together

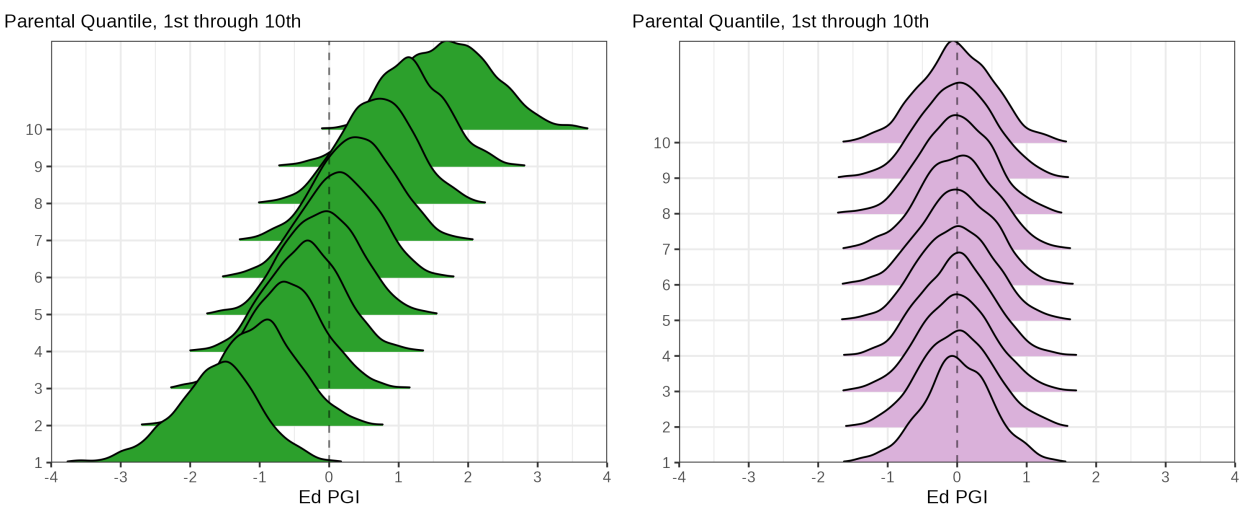
proxy for their unobserved parental PGI values.

$$\mathbb{E}\left[Z_{i,s(i)}^{\text{Parental}}\right] = \frac{1}{2}\left(Z_{m(i)}^{\text{Mother}} + Z_{f(i)}^{\text{Father}}\right) = \mathbb{E}\left[Z_i^{\text{Child}} \,\middle|\, Z_{m(i)}^{\text{Mother}}, Z_{f(i)}^{\text{Father}}\right].$$

However, because siblings collectively reveal approximately 75% of the parental genetic variation, the resulting imputed values contain measurement error compared to the ideal (but unobserved) parental mean. The sibling-imputed parental mean PGI can be viewed as a proxy for the true parental mean PGI, inflating the variance in error term u_i in Equation (1). This error reduces precision of using the imputed proxy measure, compared true parental values. Also, the coefficient between a child's Ed PGI and their expected inheritance will suffer from attenuation bias from using sibling imputed data, so is estimated in every regression in this paper (i.e., included as a control variable).

Figure 4: Distribution of the Ed PGI, Relative to Imputed Parental Decile.

(a) Ed PGI. **(b)** Ed PGI minus imputed parental mean.



Note: This figure shows the distribution of Ed PGI, Z_i^{Child}, and the Ed PGI minus imputed parental mean, $Z_i^{\text{Child}} - \mathbb{E}\left[Z_i^{\text{Child}} \,\middle|\, Z_{m(i)}^{\text{Mother}}, Z_{f(i)}^{\text{Father}}\right]$, conditional on the decile of the parental imputed Ed PGI. Parental imputed Ed PGI decile 1 shows the distributions for the bottom 10% of parental imputed values, decile 2 shows distribution among 10–20%, and so on.

Children with a higher Ed PGI value inheritance this from their parents, and so had parents with higher Ed PGI values. Figure 4 shows this tendency, where children with parents in the top decile of the imputed Ed PGI distribution have a mean Ed PGI 1.5 standard deviations above the mean — and similarly for those at the bottom. This means parental Ed PGI values, with their effects on the children and the environment they are

raised in, are a meaningful confounder for the effects of the child's own Ed PGI on education and labour market outcomes. But, after conditioning for parental imputed values, there is no correlation across the distribution; this reflects the fact that Mendelian independent assortment induces random quasi-random variation in the child's realised Ed PGI.

3.3 Causal Identification Strategy

Let $D_i(z')$ denote the potential education level and $Y_i(z', D_i(z'))$ labour market outcome for individual i, under the hypothetical alternative realisation that i had Ed PGI value $Z_i^{\text{Child}} = z'$. Observed education and wages/income data are a sum of their respective potential outcomes.

$$D_i = \sum_{z'} \mathbb{1}\left\{Z_i^{\text{Child}} = z'\right\} D_i(z'), \quad Y_i = \sum_{z'} \mathbb{1}\left\{Z_i^{\text{Child}} = z'\right\} Y_i(z', D_i(z')).$$

Potential labour market outcomes $Y_i(z', D_i(z'))$ are a function of both the Ed PGI value, and resulting education $D_i(z')$. This distinction is not important for measuring average genetic effects in this section, but will become the main focus in Section 5.

Target Parameters. This study aims to estimate the average causal effect of having a higher Ed PGI, which is summarised by the following marginal average effects:[10]

$$\text{Average education effect } \frac{\partial}{\partial z'}\mathbb{E}\left[D_i(z')\right], \quad \text{Average total effect } \frac{\partial}{\partial z'}\mathbb{E}\left[Y_i(z', D_i(z'))\right].$$

The average education effect effect represents the average effect of a one standard deviation increase in the Ed PGI on education, and the average total effect on labour market wages or income, *ceteris paribus*.

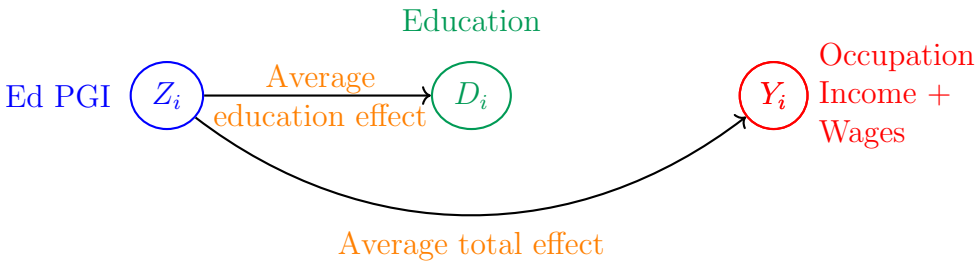
Identifying Assumptions. I rely on controlling for parental (imputed) Ed PGI values, for causal inference. This yields estimates of causal effects if a child's Ed PGI is quasi-randomly assigned, conditional on their parents' Ed PGI mean — the child's expected inheritance.

$$Z_i^{\text{Child}} \perp\!\!\!\perp D_i(z'), Y_i(z', D_i(z')) \quad \Big| \quad Z_{i,s(i)}^{\text{Parental}} \tag{2}$$

The identifying assumption — that a child's Ed PGI is quasi-randomly assigned condi-

[10] Throughout, we assume that $D_i(z')$ is differentiable in z' and $Y_i(z', D_i(z'))$ is differentiable in both arguments, together with standard regularity conditions permitting differentiation under the expectation.

Figure 5: Structural Causal Model for Ed PGI Effects.



Note: This figure shows the structural causal model, representing the two causal effects estimated in this section. The causal effects of the child's Ed PGI on education and later labour market outcomes are identified by controlling for parental (imputed) Ed PGI values.

tional on parental PGI values — is credible precisely because of the biology described above. The deviation away from expected inheritance represents the difference between a child's realised Ed PGI and the Ed PGI they were expected to inherit given their imputed parents' PGI values. This deviation arises solely from the random segregation of chromosomes during meiosis: at each of the thousands of SNPs entering the Ed PGI, whether a parent with one copy passes it on is determined by an independent coin flip, with no connection to the family environment, the child's later circumstances, or any other factor that might influence education or labour market outcomes.

This generates credible quasi-experimental genetic variation, conditional on parental PGI values. A child who drew a higher Ed PGI than their expected inheritance is not systematically different in family background, parental environment, or any other confounding factor from another who drew a lower value; the only difference is the outcome independent biological coin flips at conception. Controlling for parental PGI values therefore isolates this random component, enabling causal inference on the average effects of the Ed PGI on education and labour market outcomes.

Estimation Strategy. I estimate the average effects of Ed PGI values on education and labour market outcomes with the following regression equations.

$$D_i = \phi_1 + \pi Z_i^{\text{Child}} + \zeta_1 Z_{i,s(i)}^{\text{Parental}} + \boldsymbol{\varphi}_1' \boldsymbol{X}_i + \eta_i, \tag{3}$$

$$Y_i = \phi_2 + \theta Z_i^{\text{Child}} + \zeta_2 Z_{i,s(i)}^{\text{Parental}} + \boldsymbol{\varphi}_2' \boldsymbol{X}_i + \varepsilon_i, \tag{4}$$

where π is the effect of the Ed PGI on education variables (average education effect), θ the effect on later labour market outcomes (average total effect). Inclusion of parental imputed Ed

PGI values yields causal inference, and $\boldsymbol{X}_i$ includes a birth set of control variables including gender, sibling count, in city indicator, birth year and UKB data collection year fixed effects, and inclusion of other PGI values as controls.[11] Including these other PGI controls isolates the effects of the Ed PGI minus contribution from SNPs highly associated with other outcomes.

Using standard linear models, the marginal effects are assumed constant across the Ed PGI distribution, so that OLS consistently estimates a single parameter coinciding with the average total effect. Without linearity, OLS recovers a variance-weighted average of marginal effects, placing disproportionate weight on individuals whose Ed PGI deviates most from its conditional mean; the linearity assumption is therefore what gives the estimates their interpretation as population averages.

Limitations. Several limitations apply to the estimates presented here. First, the OLS estimates of average education effect and average total effects (Equation (3) and (4)) are not simple averages across individuals, but rather convex weighted averages of individual-level causal effects, with weights determined by the OLS projection.[12] Without assuming homogeneity of effects across the Ed PGI distribution, these estimates need not coincide with the population average total effect, and may place disproportionate weight on individuals whose Ed PGI deviates most from its conditional mean.

Second, the analysis sample is restricted to UKB participants of European ancestry who have at least one observed sibling, as described in Section 2. Internal validity is sound within this sample, although the estimated average effects may not generalise to the broader British population, whose observed and unobserved structure may systematically differ from this selected group. Third, the sibling-based imputation of parental PGI values introduces additional measurement error relative to the ideal case of directly observing true parental genotypes, attenuating the estimated coefficients. This attenuation is compounded by the fact that the Ed PGI itself is constructed from GWAS weights estimated with error, further understating the true causal effects of genetic predisposition toward educational attainment.[13]

[11] The other PGI variables are shown in Summary Table 1, including estimated PGIs for ADHD, asthma, bipolar disorder, body mass index, diabetes (type 2), height, and schizophrenia.

[12] See Angrist & Krueger (1999) and (Angrist & Pischke 2009, Section 3.5)

[13] The Ed PGI weights used in this paper (from Okbay et al. 2022) are calculated with measurement error, leading to attenuation bias relative to an infeasible truth Ed PGI with oracle weights. See van Kippersluis et al. (2023) for further discussion on measurement error of this kind.

4 Education and Total Genetic Effects

The Ed PGI has substantial causal effects on both educational attainment and labour market outcomes. Table 2 reports estimates of the average causal effects, identified by the quasi-random variation in genetic inheritance described in Section 3. Each coefficient on the Ed PGI represents the average causal effect of a one standard deviation increase in the Ed PGI, conditional on the imputed parental Ed PGI and the full set of controls. The parental Ed PGI enters as a control throughout: it is correlated with both the child's Ed PGI and the outcomes of interest, but does not represent a causal effect, as it absorbs the component of a child's Ed PGI that reflects inherited family environment rather than random genetic variation.

Table 2: Estimates of Education and Total Genetic Effects.

	Education Outcomes			Labour Market Outcomes		
Outcome:	Education years	Age left school	University completion	Occupation hourly wage	Occupation annual income	Household income (binned)
	(1)	(2)	(3)	(4)	(5)	(6)
Ed PGI	0.546	0.163	0.072	0.049	0.047	0.058
	(0.059)	(0.016)	(0.008)	(0.005)	(0.006)	(0.007)
Parental Ed PGI	0.870	0.337	0.105	0.067	0.066	0.094
	(0.078)	(0.017)	(0.012)	(0.005)	(0.006)	(0.003)
Outcome mean	13.835	16.670	0.334	2.839	3.064	3.567
Adjusted R^2	0.146	0.043	0.107	0.138	0.205	0.132
Observations	24,702	16,726	24,702	24,702	20,560	22,181

Note: This table shows estimates of the average causal effect of the Ed PGI on education variables (average education effect) and labour market outcomes (average partial effect) — specification in Equation (3) and (4). The coefficient on the Ed PGI represents the average causal effect, and coefficient on the parental Ed PGI is a control parameter. Specification for estimates in this table including gender, sibling count, in city indicator, birth year and UKB data collection year fixed effects, and inclusion of other PGI values as controls. Appendix Table A1 shows estimates varying the control sets, and with sibling fixed effects.

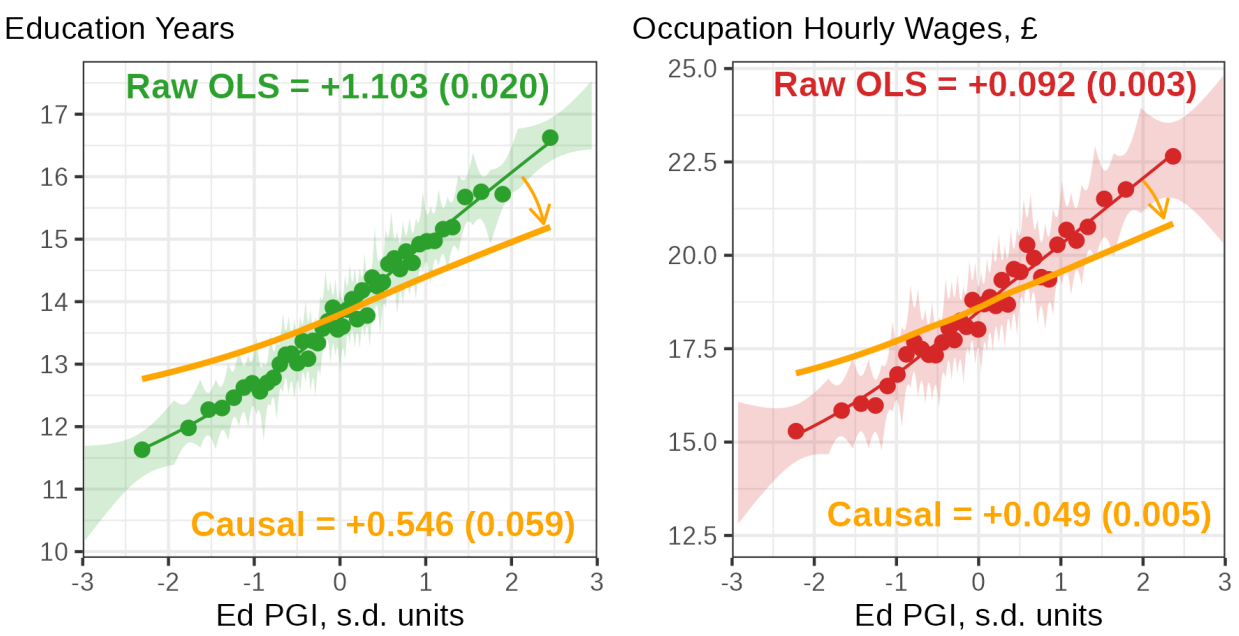
Effects on Education Variables. The Ed PGI has a meaningful causal effect on education, with the largest effects operating through formal qualification thresholds. A one standard deviation increase in the Ed PGI causes, on average, an additional 0.55 years of education (column 1; SE 0.02), against a sample mean of 13.84 years, and raises the probability of

completing university by 7 percentage points on a baseline completion rate of 33 per cent — a relative increase of around one fifth (column 3). Age of leaving full-time education, a common measure of education in the UK given that legal school-leaving age is defined by age rather than qualification, is observed for a subset of 16,728 participants; the estimated effect of 0.16 years (column 2; SE 0.02) is smaller than the qualification-based measure, reflecting that the Ed PGI operates more strongly through the decision to pursue higher qualifications than through the legally-determined school-leaving margin.

Effects on Labour Market Outcomes. The Ed PGI has a consistent and precisely estimated causal effect on earnings across all three income measures. A one standard deviation increase in the Ed PGI raises hourly wages by approximately 5 per cent (column 4) and annual occupation income by 5 per cent (column 5), with a slightly larger effect of 6 per cent on household income (column 6). These estimates capture the total average causal effect of the Ed PGI on labour market outcomes — combining any direct biological pathway with effects mediated through the additional education documented above.

Figure 6: Effect of Ed PGI on Education and Wages.

(a) Ed PGI → Education Years. **(b)** Ed PGI → Hourly Wages.



Note: This figure shows the correlation between Ed PGI and education years, as well as occupation coded hourly wages. The shifted lines are the estimates of the causal effect, estimated by controlling for parental imputed Ed PGI values.

The main estimates are robust to alternative specifications, and the main causal estimates

are half as large as raw correlation between the Ed PGI for all outcomes under study. Table A1 reports estimates under alternative specifications, and this raw correlation. Panel A shows the raw correlations between the Ed PGI and outcomes, without conditioning on parental PGI or any controls; these are roughly twice the size of the causal estimates, illustrating the extent of confounding when parental genetics are not accounted for. Panel B re-estimates the main specification without the additional control variables, and finds estimates nearly identical to those in Table 2, confirming that the causal identification rests on the parental PGI rather than the auxiliary controls. Including the other PGI for other outcomes as controls make little difference to the causal estimates, which aligns with specifications reporting correlations with the Ed PGI that do attenuate with inclusion of other PGIs (e.g., Bolyard & Savelyev 2025). Panel C replaces the parental PGI with sibling fixed effects, which partial out all shared family characteristics directly; the resulting estimates are again very close to the main results across education and labour market outcomes, lending further support to the identifying assumption that within-family variation in the Ed PGI is quasi-randomly assigned.

5 Direct and Indirect Effects

The average total effect of the Ed PGI on labour market outcomes is the sum of two conceptually distinct channels. The first is a direct channel: genetic predisposition toward education may influence earnings independently of schooling itself, through traits such as cognitive ability, non-cognitive skills, or personality characteristics that are correlated with the Ed PGI and valued by employers. The second is an indirect channel: a higher Ed PGI induces more education, and that additional education generates a labour market return.

The size and robustness of the labour market effects documented in Section 4 has led to a common interpretation in the literature that at least part of the Ed PGI's influence must operate through the direct channel. This view is often motivated by the fact that the Ed PGI predicts fluid intelligence scores and by results showing earnings associations that persist even after conditioning on education (Carvalho 2025); it appears, for example, in work arguing that the Ed PGI captures skills related to financial literacy and other non-educational attributes (Barth et al. 2020). Separating these channels matters both for understanding the biological and social mechanisms behind genetic effects on earnings, and for policy: if the indirect channel dominates, then policies that expand access to education can, in principle, replicate much of the labour market advantage associated with a higher Ed PGI; if the direct channel dominates, the policy implications are considerably more complex.

This section uses a Causal Mediation (CM) framework to decompose the average total effect of the Ed PGI into its direct and indirect components, where education years serves as the mediating variable. The key empirical challenge is that causal mediation analysis requires not only that the Ed PGI is quasi-randomly assigned (conditional on parental values), but also that the mediator is quasi-randomly assigned conditional on treatment, which is not credible for education in this setting. UKB participants were not randomly assigned education, so it takes further steps to identify the returns to an additional year of education. Based on this insight, I combine credible estimates of the average effect of the Ed PGI on education together with estimates of the returns to education to identify the direct and indirect effect channels.

5.1 Causal Mediation (CM) Framework

Target Parameters. The total average total effect of the Ed PGI on labour market outcomes can be decomposed into a direct and indirect component. Formally, the direct and indirect effects are defined as:

$$\text{Direct Effect} = \frac{\partial}{\partial z'}\mathbb{E}\left[Y_i(z', D_i(Z_i^{\text{Child}}))\right],$$
$$\text{Indirect Effect} = \frac{\partial}{\partial z'}\mathbb{E}\left[Y_i(Z_i^{\text{Child}}, D_i(z'))\right].$$

The direct effect captures the effect of the Ed PGI on earnings holding the education channel fixed; the indirect effect captures the effect that operates exclusively through the induced change in education. These sum to the average total effect θ estimated in Section 4:[14]

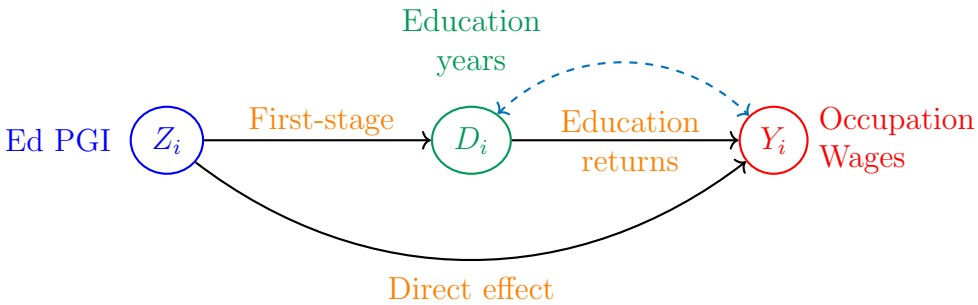
$$\underbrace{\frac{\partial}{\partial z'}\mathbb{E}\left[Y_i(z', D_i(z'))\right]}_{\text{Average total effect, }\theta} = \underbrace{\frac{\partial}{\partial z'}\mathbb{E}\left[Y_i(z', D_i(Z_i))\right]}_{\text{Direct effect}} + \underbrace{\frac{\partial}{\partial z'}\mathbb{E}\left[Y_i(Z_i, D_i(z'))\right]}_{\text{Indirect effect}},$$

evaluated at z' — again, a linearity assumption is assuming the average effect does not vary across the $Z_i^{\text{Child}} = z'$ distribution. Figure 7 illustrates this decomposition in the structural causal model: the direct effect corresponds to the Ed PGI arrow bypassing education, and the indirect effect to the path running through education years and its labour market return.

Identification. Standard CM identification requires that the mediator is quasi-randomly assigned conditional on treatment (see Imai et al. 2010). This assumption is not credible here:

[14] Appendix A.4 gives the simple derivation.

Figure 7: Structural Causal Model for Ed PGI Effects.



Note: This figure shows the structural causal model behind direct and indirect effects, where $Z_i \to D_i$ represents a causal relationship of Z_i on D_i. The dotted line between education and income & wages represents the fact that education is not randomly assigned, and a strategy for identifying returns to education is needed for valid CM estimates.

education years are shaped by family background, ability, and a host of other factors that independently influence earnings, so that the education–earnings relationship is confounded in the UKB sample even after conditioning on the Ed PGI. However, it is possible to identify the direct and indirect effects with a valid estimate for the causal returns to education, by the chain rule.[15]

$$\text{Indirect Effect} = \underbrace{\frac{\partial}{\partial z'}\mathbb{E}\left[D_i(z')\right]}_{\text{Average first-stage, } \pi} \times \underbrace{\frac{\partial}{\partial d'}\mathbb{E}\left[Y_i(Z_i, d')\right]}_{\text{Returns to education, } \beta}.$$

The indirect effect equals the average first-stage effect π — credibly identified by the Mendelian design — multiplied by the returns to an additional year of education β. The direct effect then follows as average total effect minus the indirect effect,

$$\text{Direct Effect} = \theta - \pi\beta.$$

Estimation. I estimate the mediation second-stage from the following specification:

$$Y_i = \alpha + \beta D_i + \gamma Z_i^{\text{Child}} + \delta Z_i^{\text{Child}} D_i + \zeta Z_{i,s(i)}^{\text{Parental}} + \boldsymbol{\varphi}' \boldsymbol{X}_i + \varepsilon_i, \tag{5}$$

where β captures average returns to education, γ the residual direct effect of the Ed PGI on earnings conditional on education, and δ the interaction between the Ed PGI and education years. Since Z_i^{Child} is normalised to have mean zero, the interaction term $\delta Z_i D_i$ has mean

[15]This additionally requires the smoothness and regularity conditions assumed in Section 3.

zero across the sample, so that β recovers the average returns to education evaluated at the population mean of the Ed PGI.[16]

Limitations. Several limitations apply to the CM estimates presented in this section. First, the OLS estimates of the average total effect and average first-stage are variance-weighted averages of individual-level effects under heterogeneity (as noted in Section 3). Without the linearity assumption, these average effects need not coincide with population average effects. Second, and more substantively, the indirect effect in Section 5.1 implicitly assumes that the returns to education are homogeneous: individuals induced into additional schooling by a higher Ed PGI face the same average returns as the broader population. In the language of CM, this is an assumption about the education compliers — those whose schooling decisions are shifted by the Ed PGI — whose returns may differ from the population average if, for example, education-linked genes select individuals into different schooling margins with different labour market rewards. The validity of the CM estimates therefore rests on the credibility of β as an estimate of the returns to education relevant for this complier population. What remains, therefore, is to provide credible estimates of the causal returns to education in the UKB sample.

5.2 Returns to Education

Correlational estimates of education returns are the most natural starting point. A large literature finds that OLS estimates of the return to an additional year of schooling are remarkably stable — typically in the range of 5–10 per cent in log earnings — and, perhaps surprisingly, close to estimates from quasi-experimental designs exploiting compulsory schooling reforms (Angrist & Krueger 1991, Harmon & Walker 1995, Oreopoulos 2006). Two forces likely explain this convergence: ability bias pushes OLS upward, while measurement error in reported schooling pushes it downward, and the two roughly cancel. Quasi-experimental designs, meanwhile, often identify effects among specific groups of marginal compliers — those induced into extra schooling by a particular reform — whose returns need not equal the population average (Buscha & Dickson 2015). That OLS and quasi-experimental estimates land in a similar range is therefore informative, not coincidental.

Table 3 shows that estimates for the labour market returns to education in the UKB

[16] Hogan-Hennessy (2025) shows that the interaction term δ does not suffer from selection bias even when the mediator is not randomly assigned, provided the treatment is quasi-randomly assigned — a condition satisfied here by the Mendelian design.

Table 3: Returns to Education, Correlational Estimates.

Outcome:	Occupation hourly wage			Occupation annual income			Household income (midpoint imputed)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Education years	0.061	0.059	0.059	0.061	0.059	0.059	0.068	0.063	0.063
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)
Ed PGI		0.010	0.000		0.008	-0.001		0.008	-0.015
		(0.011)	(0.010)		(0.021)	(0.021)		(0.013)	(0.010)
Education years × Ed PGI		0.001	0.001		0.001	0.001		0.003	0.003
		(0.001)	(0.001)		(0.001)	(0.001)		(0.001)	(0.001)
Parental Ed PGI			0.016			0.016			0.040
			(0.003)			(0.005)			(0.007)
Collected Education Returns	0.061	0.059	0.059	0.061	0.059	0.059	0.068	0.064	0.063
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)
R^2	0.328	0.332	0.332	0.273	0.274	0.274	0.204	0.208	0.208
Observation count	24,702	24,702	24,702	20,560	20,560	20,560	22,181	22,181	22,181

Note: This table shows simple OLS estimates for returns to education in the UK Biobank sibling sample — the relevant sample for causal genetic effects. All outcomes are in a log specification, and each column is a regression including the controls previous mentioned, with specification in Equation (5), varying inclusion of the Ed PGI and parental Ed PGI as indicated. Observation counts vary because each outcome has a different number of observations which are non-missing. The final row for education returns collects the partial effect of education years, including the interaction terms between education years and the Ed PGI. Point estimates are relatively similar to those using the MLSA rise as an instrument (see Appendix A.5), with much more precision.

sample are no exception. Across all three income measures, the implied return to an additional year of education sits consistently at around 6 per cent in log earnings. This estimate is stable across specifications: adding controls for the Ed PGI and its interaction with education years barely moves the point estimates, confirming that the return does not simply reflect residual genetic confounding; including the parental Ed PGI changes nothing. The UKB correlational estimates are, in short, consistent with the broader literature.

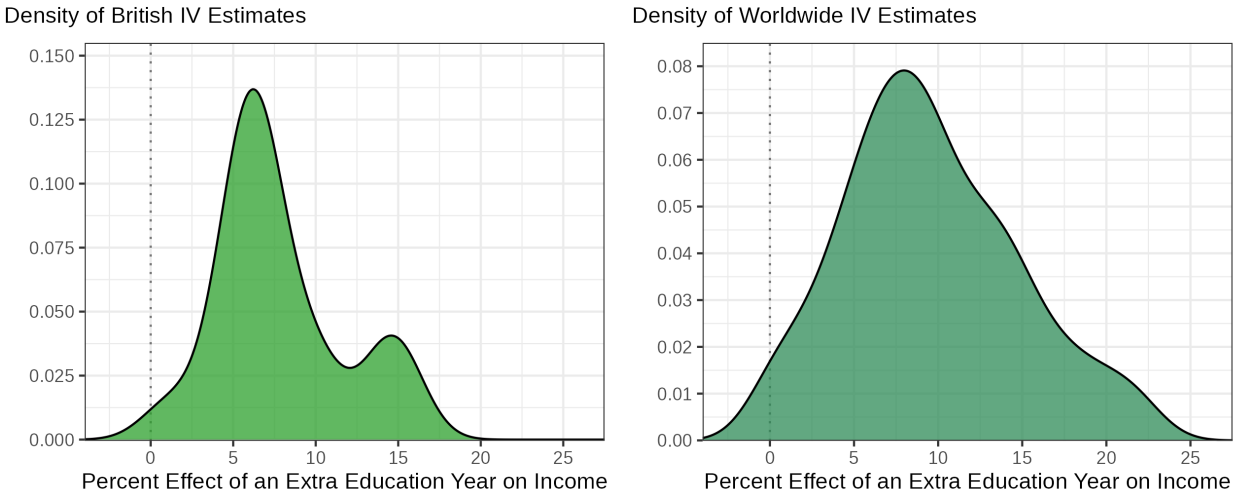
Education years are not randomly assigned, and no correlational estimate is fully credible on its own. The UKB does contain a plausible quasi-experiment: the 1972 rise in the British Minimum School Leaving Age (MLSA), which compelled children born after September 1957 to stay in school until age 16. While this sample lines up well with the UKB population, the resulting fuzzy RDD estimates of education returns (in the range of 4–6 per cent in log earnings) carry standard errors too large to rule out either zero or substantially larger

effects, and are therefore too imprecise for use in the mediation analysis.[17] Given this, the correlational estimates anchored in the broader UK literature offer a more useful basis for the mediation analysis, and the sensitivity analysis below makes the dependence on any particular return value transparent.

5.3 Sensitivity Analysis

The indirect effect is the product of two quantities: the effect on education years and the return to an additional year of education. Since these multiply together, any uncertainty in the education return translates directly and proportionally into uncertainty about the share of the total genetic effect that operates through education. Rather than conditioning the mediation analysis on a single return value, I vary it across its plausible range. In addition, this range is defined not arbitrarily, but based on the previous labour economics on education returns.

Figure 8: Distribution of Economics Literature for Income Returns to Education.



Note: These figures show the estimated kernel density for the distribution of estimates for later-life income returns to education. Panel A refers to the implied distribution from 17 estimates using IVs in the UK (shown in full in Appendix Table A3), and Panel B for 182 estimates from 144 studies across the world. These estimates were collected by Patrinos & Psacharopoulos (2025).

The return to an extra year of education in the UK has been estimated many times in the economics literature, giving a rich empirical base for calibrating the sensitivity analysis. Figure 8 shows the distribution of 17 quasi-experimental UK estimates collected by Patrinos

[17] Appendix A.5 details estimates for labour market education returns using the MLSA rise as a regression discontinuity, as previously considered by Barcellos et al. (2021), Carvalho (2025).

& Psacharopoulos (2025), listed in full in Appendix Table A3. The distribution is right-skewed: most estimates fall between 4 and 8 per cent, consistent with the correlational estimates in Table 3, but a tail of larger estimates reaches to around 15 per cent. These higher values tend to come from studies exploiting the 1947 compulsory schooling reform, which identifies returns among a different and older complier population than the UKB cohort. The worldwide distribution from 182 estimates across 144 studies, shown alongside, is broadly consistent with the UK evidence.

The first step of the sensitivity analysis asks a deliberately agnostic question: how do the mediation estimates change as the assumed education return varies across the full span of values that appear in the literature, from -5 to 20 percent? For each value in this range, the indirect effect follows directly as the product of the estimated genetic first-stage and the assumed return, with the direct effect as the remainder. This step takes no stand on which return value is correct, and shows whether the mediation conclusions are stable or fragile across the plausible range.

The second step uses the empirical UK distribution more directly. I bootstrap across the 17 IV estimates in Table A3, drawing a return value at each replication and computing the implied mediation decomposition. Repeating this across many draws produces a distribution of indirect and direct effect estimates that reflects both sampling uncertainty in the effect of the Ed PGI on education years and the genuine disagreement in the literature about the magnitude of education returns. Reporting quantiles of this distribution gives a meta-analytic sensitivity bound anchored in what the literature actually finds, rather than treating all values across the plausible range as equally likely.

Together, the two steps answer different questions. The first shows the full range of possible conclusions. The second shows where the weight of evidence places them.

6 Direct and Indirect Effect Estimates

Long section describing the mediation estimates.

Use estimate the mediation effects, using (1) OLS (2) sibling FEs (3) MSLA.

7 Discussion and Future Directions

Discussion goes here.

Table 4: Direct and Indirect Effect Estimates, using Correlational Education Returns.

Outcome:	Occupation hourly wage		Occupation annual income		Household income (binned)	
	(1)	(2)	(3)	(4)	(5)	(6)
Education Effect	0.565	0.551	0.572	0.557	0.569	0.559
	(0.037)	(0.037)	(0.040)	(0.040)	(0.039)	(0.039)
Total Genetic Effect	0.050	0.049	0.042	0.048	0.056	0.059
	(0.005)	(0.005)	(0.009)	(0.008)	(0.009)	(0.008)
Education Returns	0.058	0.059	0.064	0.060	0.071	0.064
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Direct, Ed PGI effect	0.018	0.017	0.006	0.015	0.016	0.023
	(0.004)	(0.004)	(0.009)	(0.008)	(0.009)	(0.008)
Indirect, Education Effect	0.033	0.033	0.036	0.033	0.041	0.036
	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)
Percent mediated through education	0.651	0.653	0.853	0.693	0.720	0.611
	(0.056)	(0.059)	(0.224)	(0.129)	(0.128)	(0.091)
Controls included?	No	Yes	No	Yes	No	Yes
Observation count	24,743	24,702	20,592	20,560	22,215	22,181

Note: This table shows the mediation point estimates, of the effect of the Ed PGI going through education years (indirect effect) and a direct genetic effect. Specification controls for parental Ed PGI and uses correlation returns to education, as mentioned in the previous section. All outcomes are in a log specification, and each column varies inclusion of the control variables as indicated, and standard errors on the mediation estimates are calculated by the bootstrap with 10,000 bootstrap samples. Observation counts vary because each outcome has a different number of observations which are non-missing.

8 Summary and Concluding Remarks

Conclusion goes here.

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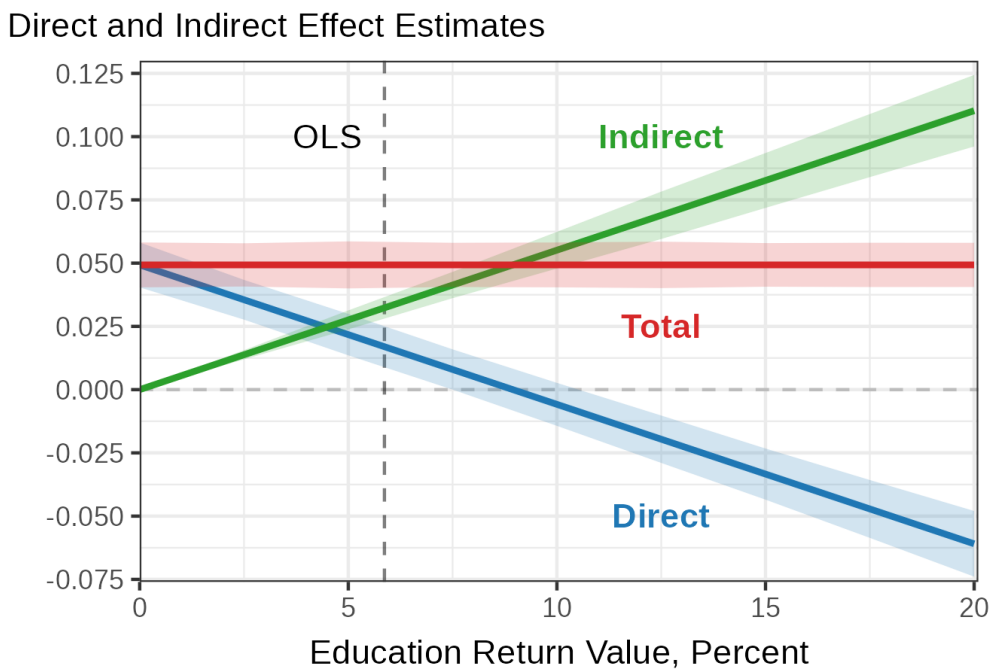
Figure 9: Sensitivity Analysis, Effect of Ed PGI on Log Occupational Wages via Education Years.



Note: This figure shows how the indirect (education) and direct effect estimates vary using difference values for average returns to education. The total effect refers to the average effect of the Ed PGI on log occupational hourly wages after controlling for parental Ed PGI values. The shaded regions refer to 95% confidence intervals, calculated with bootstrapped standard errors. The dotted line labelled "OLS" shows the position of the correlational returns to education estimate, presented in Table 3. The right

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A Supplementary Appendix

This project used computational tools which are fully open-source. Any comments or suggestions may be sent to me at seh325@cornell.edu, or raised as an issue on the Github project.

A number of statistical packages, for the R language (R Core Team 2023), made the empirical analysis for this paper possible.

- *Tidyverse* (Wickham et al. 2019) collected tools for data analysis in the R language.

A.1 Occupational Imputed Wages and Income

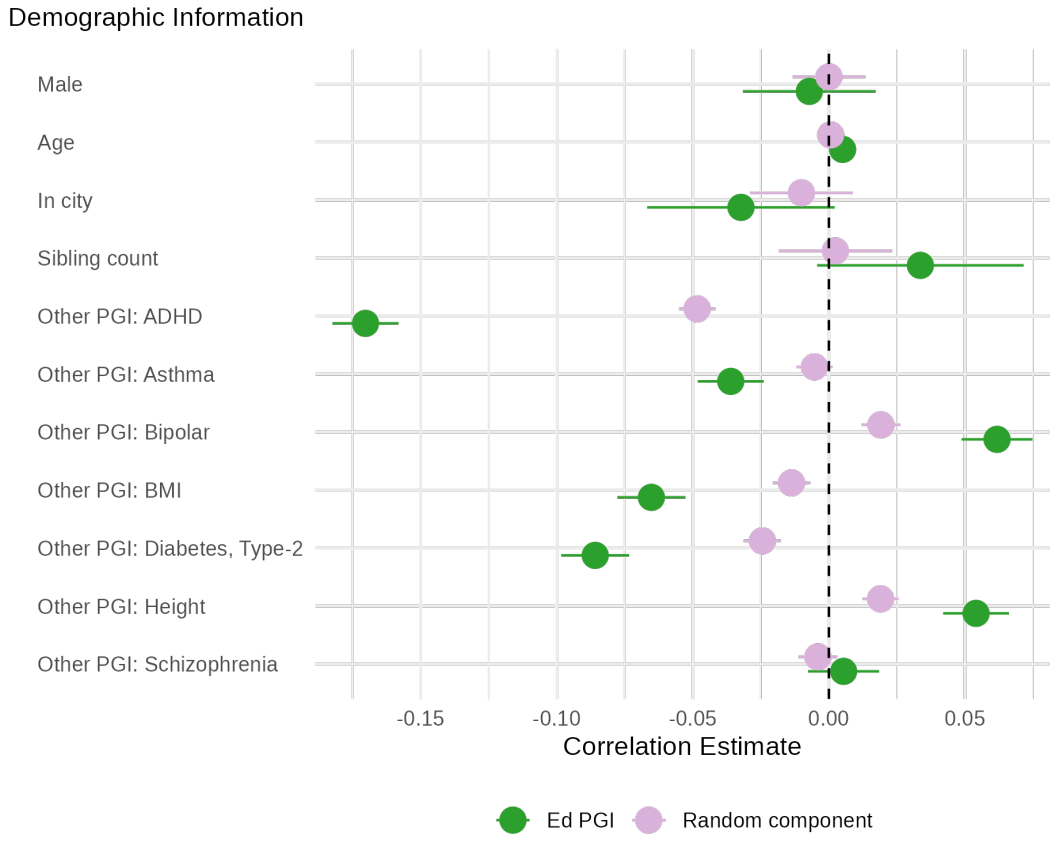
Describe here the Kweon Koellinger imputation procedure. Reference their supplementary data, and thanks for additional guidance on it.

A.2 Identifying Family Clusters by Genetic Similarity

Describe here.

A.3 Genetic Effects

Figure A1: Correlation for Ed PGI, and Ed PGI Minus Parental Mean, with Demographic Variables.



Note: Multivariate regression estimates for the linear model $Z_i = \boldsymbol{\phi}' \boldsymbol{X}_i + \eta_i$, where $\boldsymbol{X}_i$ is the vector demographic variables described by the labels on the y-axis. The dots are the point estimates, and bars are the 95% confidence intervals.

Table A1: Different Specifications for Estimates of Average Genetic Effects.

	Education Outcomes			Labour Market Outcomes		
Outcome:	Education years	Age left school	University completion	Occupation hourly wage	Occupation annual income	Household income
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Raw Correlation						
Ed PGI	1.103	0.361	0.140	0.092	0.081	0.109
	(0.020)	(0.020)	(0.003)	(0.003)	(0.005)	(0.005)
Panel B: Including Parental Ed PGI with no Control Variables						
Ed PGI	0.565	0.161	0.074	0.050	0.042	0.056
	(0.037)	(0.035)	(0.005)	(0.005)	(0.009)	(0.009)
Parental Ed PGI	0.855	0.327	0.105	0.067	0.062	0.084
	(0.050)	(0.047)	(0.007)	(0.006)	(0.012)	(0.012)
Panel C: Including Family Fixed Effects						
Ed PGI	0.560	0.294	0.070	0.052	0.061	0.065
	(0.039)	(0.053)	(0.006)	(0.005)	(0.010)	(0.010)
Outcome mean	13.835	16.670	0.334	2.839	3.064	3.567
Observations	24,702	16,726	24,702	24,702	20,560	22,181

Note: This table shows estimates of the average causal effect of the Ed PGI on education variables (average first-stage) and labour market outcomes (average partial effect) — specification in Equation (3) and (4). The coefficient on the Ed PGI represents the average causal effect, and coefficient on the parental Ed PGI is a control parameter. Specification s vary for each Panel, as described.

A.4 Causal Mediation (CM) Decomposition

The decomposition of the average partial effect into direct and indirect components follows from a straightforward application of the multivariate chain rule. Define:

$$g(a, b) = \mathbb{E}\left[Y_i(a, D_i(b))\right].$$

The average partial effect, direct effect, and indirect effect can then be written as:

$$\begin{aligned}
\text{Average partial effect} &= \frac{d}{dz'} g(z', z'), \\
\text{Direct effect} &= \frac{\partial}{\partial a} g(a, Z_i)\Big|_{a=z'}, \\
\text{Indirect effect} &= \frac{\partial}{\partial b} g(Z_i, b)\Big|_{b=z'}.
\end{aligned}$$

The average partial effect moves both arguments of g simultaneously with z', whereas the direct and indirect effects move only the first and second arguments respectively, holding the other fixed. By the multivariate chain rule:

$$\frac{d}{dz'}g(z', z') = \frac{\partial g}{\partial a}(z', z') + \frac{\partial g}{\partial b}(z', z'),$$

which establishes that the average partial effect is exactly the sum of the direct and indirect effects, evaluated at $z' = Z_i^{\text{Child}}$. This result is exact under the smoothness and regularity conditions assumed in Section 3, with no further approximation required.

A.5 Returns to Education by the MLSA Rise

The UK Biobank includes a total of 500,000 members of the British public, with a roughly even distribution of birth years over 1945–1965. In 1972, the British government increased the Minimum School Leaving Age (MSLA) by one year, from age 15 to 16. The change took effect at the start of the 1972-73 school year, and applied to those born on or after 1 September 1957. The reform therefore induced extra school years for individuals born around date 1 September 1957. British school children sit exams for the GCSE qualification at age 16, so the reform pushed a larger share of the UK Biobank cohorts to remain in school to 16 and to complete these qualifications.

The 1972 MLSA change has been previously used for labour market education returns (Oreopoulos 2006), and (null) mortality effects of compulsory education (Clark & Royer 2013). The UK Biobank population overlaps well with the birth date cutoff, and so has been used in a series of papers for estimating education returns in the UK Biobank, for health outcomes (Barcellos et al. 2023, 2025, Davies et al. 2018) and occupation imputed income (Barcellos et al. 2021, Carvalho 2025).

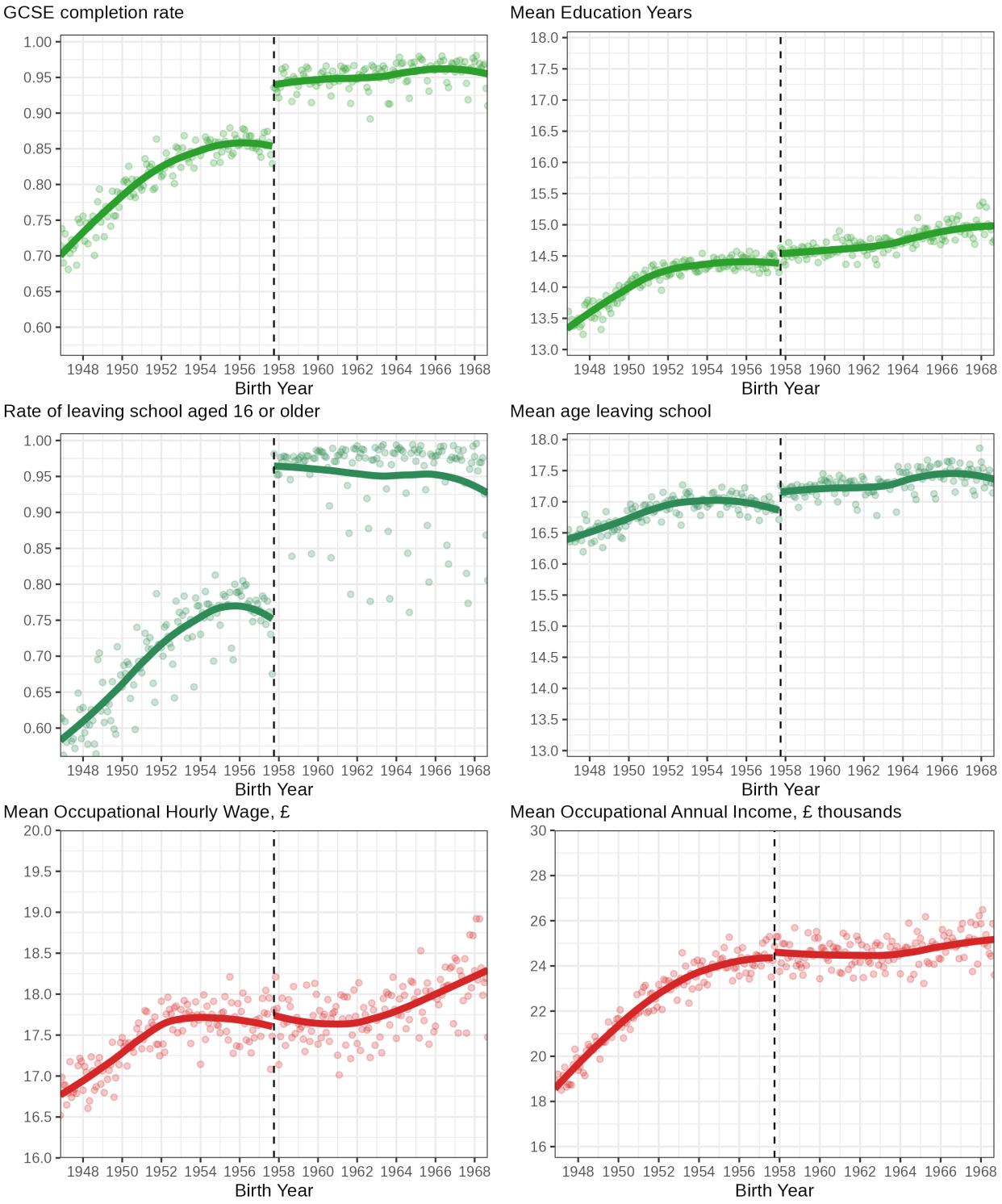
Birth month is the running variable in a fuzzy Regression Discontinuity Design (RDD), using the indicator for being born September 1957 and after as an instrument. Because not all those eligible necessarily stayed beyond 15, the post-cutoff indicator instruments education variables, where mentioned. I estimate the following RDD estimands, estimating the effect of the MSLA change on education variables — represented by D_i.

$$\lim_{v' \to^- 1957.75} \mathbb{E}\left[D_i \,|\, V_i = v'\right], \qquad \lim_{v' \to^+ 1957.75} \mathbb{E}\left[D_i \,|\, V_i = v'\right]$$

where V_i represents individual i's birth year plus birth month divided by 12 (so that 1957.75 represents born in September 1957), and $\to^-$ refers to the limit from below, $\to^+$ above. The goal is to identify is the average compliance rate of the MSLA rise,

$$\mathbb{E}\left[D_i(1) - D_i(0)\right] = \lim_{v' \to^+ 1957.75} \mathbb{E}\left[D_i \,|\, V_i = v'\right] - \lim_{v' \to^- 1957.75} \mathbb{E}\left[D_i \,|\, V_i = v'\right].$$

Figure A2: Mean Effects of the MSLA Rise.



Note: These figures show the effects of the 1972 MSLA rise on multiple outcomes, where individuals born after September 1957 (the dashed line) were legally compelled to stay in school until age 16, whereas before only until age 15. Each dot is the mean for UK Biobank respondents with non-missing education data born in that month-year, and the lines are local polynomials either side of September 1957.

The MLSA rise significantly increases education years, completing of age 16 secondary school qualifications (GCSEs) — these are measured of education coded by qualifications, as used by Okbay et al. (2022) in calculating the Ed PGI. It had an even larger effect on age at leaving school, which are the education variables that Barcellos et al. (2025), Carvalho (2025) focus on. These raw means are shown in Figure A2. Note that the follow-up null estimates for education returns are argued to be a result of the MLSA rise inducing a low-quality margin of education Clark & Royer (2013), so while age leaving education variables are more significant for this IV they may be economically relevant.

I also use the RDD design to measure changes in later labour market outcomes. The UK Biobank asks questions of the sector people work in, giving high detail occupation codes (i.e., SOC codes) Using this value, and other demographic details, I impute occupation coded income for the UK Biobank respondents using data from the British Annual Survey of Hours and Earnings.[18] This gives occupation coded hourly income, and annual wage calculated from hours worked as outcome measures in the MLSA rise,

$$\mathbb{E}\left[Y_i(1) - Y_i(0)\right] = \lim_{z' \to {}^{+}1957.75} \mathbb{E}\left[Y_i \mid Z_i = z'\right] - \lim_{z' \to {}^{-}1957.75} \mathbb{E}\left[Y_i \mid Z_i = z'\right].$$

The final panels of Figure A2 show these outcomes, and this reduced-form effect of the MLSA is not significantly different from zero. Barcellos et al. (2021) analyse this RDD with UKB data, and a hard coded RDD specification; this subsection returns to this RDD and finds that returns to education estimates among the UKB population (or any subsample) are not robust to RDD best practices.

[18]Thanks to Kweon et al. (2025) for sharing code on this imputation.

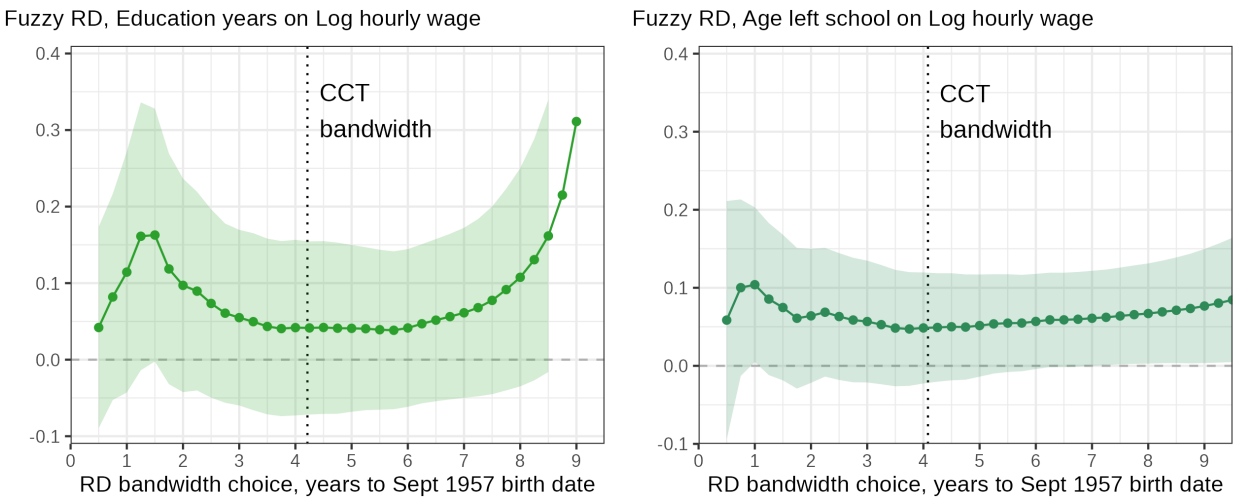
Table A2: Estimates of the MSLA Change Effects.

Panel A: First-stage.	First-stage.				Reduced form.			
Outcome:	GCSE completion		Education years		Hourly wage		Annual income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
MSLA Effect	0.089	0.247	0.158	0.314	0.006	0.006	0.010	0.010
	(0.006)	(0.011)	(0.060)	(0.058)	(0.007)	(0.007)	(0.010)	(0.010)
F-statistics	216	530	6.85	29.7	0.939	0.939	0.830	0.830
CCT bandwidth	2.66	1.60	3.21	2.60	4.10	4.10	3.87	3.87
Bandwidth observations	64,011	24,185	77,881	38,718	99,646	99,646	93,885	93,885
Panel B: Second-stage.	OLS.				MLSA IV.			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Outcome: Log Occupation Hourly Wage								
GCSE completion	0.372	0.193			0.070	0.068		
	(0.008)	(0.008)			(0.094)	(0.039)		
Education years			0.061	0.034			0.044	0.057
			(0.000)	(0.002)			(0.047)	(0.029)
CCT bandwidth	10.0	10.0	10.0	10.0	2.54	2.71	4.21	4.08
Bandwidth observations	245,740	149,732	245,740	149,732	62,027	39,938	101,570	58,987
Outcome: Log Occupation Annual Income								
GCSE completion	0.392	0.188			0.100	0.046		
	(0.015)	(0.009)			(0.151)	(0.065)		
Education years			0.065	0.032			0.062	0.048
			(0.001)	(0.002)			(0.080)	(0.049)
CCT bandwidth	10.0	10.0	10.0	10.0	2.61	2.86	3.84	4.16
Bandwidth observations	245,740	149,732	245,740	149,732	64,011	42,307	93,885	60,155
Qualification definition	Yes		Yes		Yes		Yes	
School age definition		Yes		Yes		Yes		Yes

Note: These point estimates are calculated among all UK Biobank participants born within a bandwidth of the MLSA rise cutoff (not just the sibling subsample). The odd columns use a definition of education attainment (age 16 schooling) based on reported education qualifications; even columns report based on school leaving age. The Panel B OLS results show estimates of an OLS specification for the correlation between education and wages/income with local polynomial trends each side of 1957.75 birthdate, attempting to mimic an RD set-up with a hard-coded bandwidth of 10 years for comparison.

I estimate this RDD equation with triangular kernel weights (weighting closer to the cutoff), an optimal bandwidth (Calonico et al. 2014, 2015), and non-parametric controls on either side of the cutoff (Gelman & Imbens 2019). These estimates are collected in Table A2.

Figure A3: Fuzzy RDD Estimates, Sensitivity to Bandwidth Choice.



Note: These figures show the sensitivity of fuzzy RD estimates using the MLSA discontinuity as an IV for education years (qualification measure) or age leaving school, using a different bandwidth choice for each. The shaded regions are the 95% confidence interval. The dotted line with CTT refers to the Calonico et al. (2014) optimal bandwidth choice, calculated using *rdrobust* in *R* (Calonico et al. 2015).

The MLSA rise increased years of education, as measured in both qualification definitions and school leaving age definition. However, the outcome of occupation coded wages and annual income are not strong; correspondingly, the IV estimates using the MLSA rise for education returns are not distinguishable from zero. This holds true for a range of bandwidth choices, shown in Figure A3.

These results contrast with those in (Barcellos et al. 2021, Carvalho 2025), who do not use these best practices for the 1957 cohort MLSA rise with UK Biobank data. Instead, use of a hard-coded bandwidth of 10 years and a hardcoded polynomial order 1 or 2. RDD specifications using this specification show highly efficient estimates of the education returns in the UKB sibling sample — results which do not survive RDD best practices.

A.6 Returns to Education Literature

Table A3: Returns to Education, British Estimates from the Literature.

OLS	IV	Year	Reference
29.2	5.7	2010	Dickson, M. and Smith, S., 2011. What determines the return to education: an extra year or a hurdle cleared?. Economics of education review, 30(6), pp.1167-1176.
12.2	6.6	2009	Devereux, P.J. and Fan, W., 2011. Earnings returns to the British education expansion. Economics of Education Review, 30(6), pp.1153-1166.
9.6	5.3	2009	Devereux, P.J. and Fan, W., 2011. Earnings returns to the British education expansion. Economics of Education Review, 30(6), pp.1153-1166.
9.5	6.9	2006	Grenet, J. 2013. Is it enough to increase compulsory education to raise earnings? Evidence from French and British compulsory schooling laws. Scandinavian Journal of Economics 115: 176-210.
7.8	6.2	2009	Devereux, P.J. and Fan, W., 2011. Earnings returns to the British education expansion. Economics of Education Review, 30(6), pp.1153-1166.
6.0	4.4	1999	Blanchflower, D. and Elias, P., 1999. Ability, schooling and earnings: Are twins different?. Unpublished manuscript, Warwick University.
6.9	6.6	2009	Devereux, P.J. and Fan, W., 2011. Earnings returns to the British education expansion. Economics of Education Review, 30(6), pp.1153-1166.
7.0	7.0	2001	Devereux, P.J. and R.A. Hart. 2010. Forced to be Rich? Returns to Compulsory Schooling in Britain. Economic Journal 120(549): 1345-64.
7.3	7.7	1999	Bonjour, D., Cherkas, L.F., Haskel, J.E., Hawkes, D.D. and Spector, T.D., 2003. Returns to education: Evidence from UK twins. American Economic Review, 93(5), pp.1799-1812.
4.8	5.5	1991	Dearden, Lorraine. 1999. Qualifications and earnings in Britain: how reliable are conventional OLS estimates of the returns to education? IFS Working Papers W99/07, Institute for Fiscal Studies.
4.8	5.5	1981	Dearden, L. (1998), "Ability, Families, Education and Earnings in Britain," Institute for Fiscal Studies Working Paper No. W98/14.
0.0	1.2	1947	Chib, Siddhartha, and Liana Jacobi. 2016. "Bayesian Fuzzy Regression Discontinuity Analysis and Returns to Compulsory Schooling." Journal of Applied Econometrics 31: 1026- 1047.
5.0	9.9	1970s	Harmon, C. and Walker, I. (2000), "Returns to the Quantity and Quality of Education: Evidence for Men in England and Wales," Economica, 67: 19-35.
4.6	10.2	1972	Dickson, Matt. 2013. "The Causal Effect of Education on Wages Revisited." Oxford Bulletin of Economics and Statistics 75 (4): 477-498.
7.8	14.7	1947	Oreopoulos, P. 2006. Estimating Average and Local Average Treatment Effects of Education when Compulsory Schooling Laws Really Matter American Economic Review96(1) 152-175
4.9	14.0	1992	Harmon, C. and Walker, I. (1999), "The Marginal and Average Return to Schooling in the UK," European Economic Review, 43(4-6), pp.879-887.
6.1	15.3	1947	Harmon, C. and I. Walker. 1995. Estimates of the economic return to schooling for the United Kingdom. American Economic Review 85: 1278-86.

Note: This table shows the education returns estimates on later-life labour market earnings in the economics literature, using an IV to causally estimate the effect of a year of education. Literature review kindly provided by Patrinos & Psacharopoulos (2025).